Role of AGN in Galaxy Mergers and Interactions

Brett Renaud, Jeyhan Kartaltepe

School of Physics and Astronomy, Rochester Institute of Technology, Rochester New York

Introduction

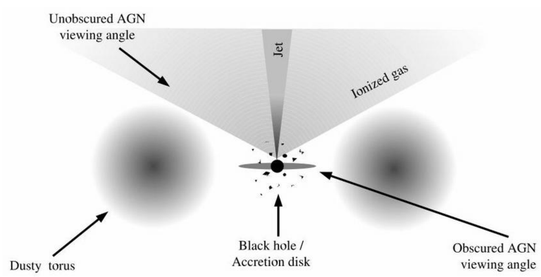
Galaxy mergers play a major role in shaping galaxy evolution, driving both star formation and the growth of supermassive black holes.

However, the **timing and strength** of the connection between **Active Galactic Nuclei (AGN)** and **merger stage** remain unclear.

Research Question:

How does AGN activity correlate with different stages of galaxy mergers across redshifts 0–3?

Structure of an Active Galactic Nucleus (AGN):



- AGN = accreting supermassive black holes.
- Galaxy mergers can drive gas inflows that fuel AGN.
- COSMOS-Web with JWST allows high-resolution studies of this connection.

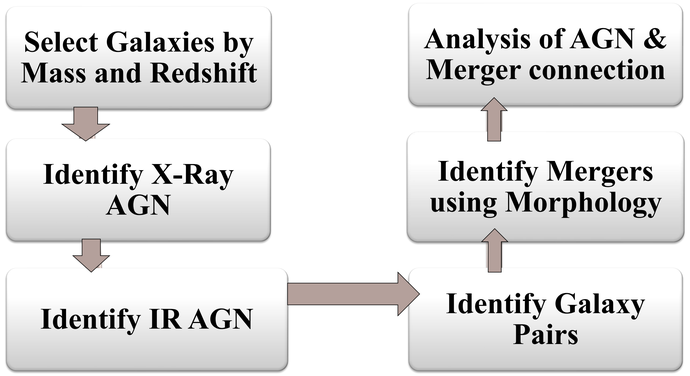
Data & Methodology

Overview & Metrics:

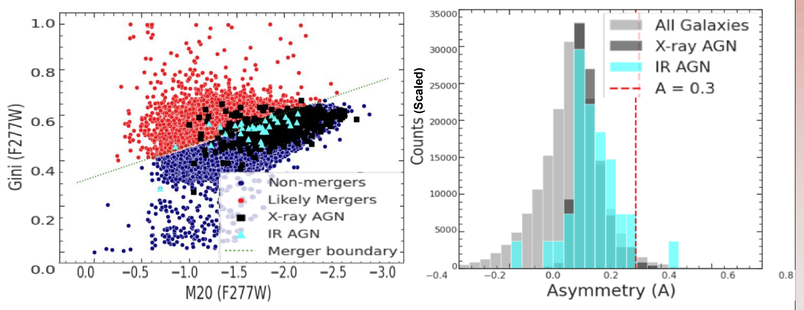
- JWST COSMOS-Web: morphology, redshift, stellar mass.
- Chandra X-ray: AGN identification by luminosity.
- Spitzer IRAC: mid-infrared AGN selection (Donley et al. 2012).

Gini: Flux concentration across pixels	M_{20}: Spatial extent of brightest 20% of light	Asymmetry: Symmetry of galaxy under 180° rotation

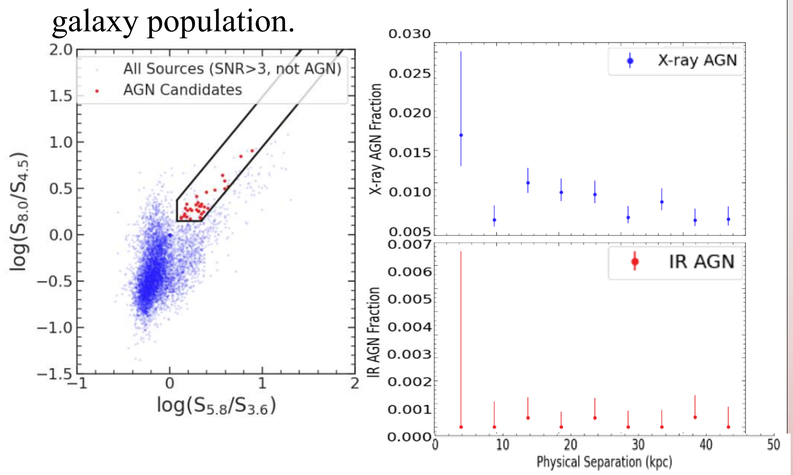
Workflow:



Results & Discussion



- AGN hosts cluster near the Gini–M20 merger boundary.
- AGN hosts show higher asymmetry than the general galaxy population.



- AGN occupy the high-S/N, AGN-candidate region of the IRAC color space.
- IR AGN fraction peaks at <10 kpc separations, consistent with merger-driven fueling.

Acknowledgements

I would like to thank my research mentor, Dr. Jeyhan Kartaltepe, as well as the Collegiate Science and Technology Entry Program(CSTEP) for their guidance and support during the summer research program.

Conclusions

- AGN are more common in morphologically disturbed galaxies, indicating a link to mergers.
- IR AGN peak at the smallest separations, supporting merger-driven inflows and obscuration.
- X-ray and IR AGN highlight different evolutionary phases of merging systems.
- JWST's high-resolution imaging enables new insights into AGN triggering from z=0–3.

References

Lotz et al. 2004, AJ, 128, 163; Donley et al. 2012, ApJ, 748, 142; Casey et al. 2023, ApJ, 954, 31; Tanmaya Mishra, 09 2016.